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Evaluation of hybrid lattice Boltzmann models for laminar and turbulent natural convection

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A R T I C L E I N F O

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A B S T R A C T

The paper analyzes different hybrid lattice Boltzmann models in buoyancy driven flow problems. The single relaxation time (BGK), two relaxation time (TRT) and high order regularized (RLB) collision models coupled with the first (1), second (2) and fourth (4) order approximation of energy equation were studied when modelling natural convection in a closed rectangular cavity. Laminar, laminar-to-turbulent and developed turbulent flows with the Rayleigh number (Ra) up to 10^{12} were considered. The local, mean and turbulent characteristics were validated against benchmark direct numerical simulation data of other researchers. It was found that all hybrid modes under consideration predicted the same results with $Ra \leq 10^8$. The TRT-2 scheme had a satisfactory performance within the laminar-to-turbulent flow behavior in a range of $6.4{\cdot}10^8 \leq Ra \leq 10^{10}$. On the other hand, the RLB-4 model demonstrated an outstanding numerical stability within developed thermal turbulence region. In particular, this scheme was stable even with $Ra = 10^{12}$.

1. Introduction

Thermal turbulence is one of the fundamental mechanisms of heat transfer. This type of heat and mass transport receives an immense attention of scientific community due to its wide-spreading in many engineering applications such as HVAC and energy storage systems, electronics cooling and many others. Despite a noticeable progress in numerical simulation of laminar natural convection, an accurate prediction of buoyancy-driven turbulence is still a great challenge. Conventional numerical techniques based on Navier-Stokes equations have some restrictions in terms of turbulence simulation. Direct numerical simulation (DNS) [1–3] and large eddy simulation (LES) [4,5] approaches are still computationally expensive when considering developed turbulent flow. On the other hand, Reynolds-averaged Navier-Stokes [6–8] method suppresses fundamental features of thermal turbulence due to high production rate of turbulent viscosity.

Past two decades a mesoscopic numerical approach called the lattice Boltzmann method (LBM) has been extensively developed. This approach [9,10] is considered as an alternative computational fluid dynamics (CFD) technique which uses particles probability distribution function to recover macroscopic quantities of the fluid. Scientific community pays a considerable attention to the lattice Boltzmann method due to its relative simplicity and suitability for massive parallel computing. With this regard, the LBM is a good candidate for no model simulation of thermal turbulence. However, the well-known problem of the LBM is numerical instability with high Rayleigh number flows. In order to overcome this weakness, some models and algorithm modifications were proposed. The most relevant studies are discussed below.

Buoyancy-driven turbulence is usually studied in terms of a closed square cavity with side heating/cooling. It needs to be stressed that the most common lattice Bhatnagar-Gross-Krook (LBGK) scheme with double population thermal model is stable with $Ra \leq 10^8$. Further increase in the buoyancy force leads to numerical instabilities and special treatment is needed for stable simulations. Somehow, this simple model was stabilized with $Ra = 10^9$ in [11–13]. Moreover, Alinejad and Isfahani [12] showed that their model is stable even with 100×100 grid points. In order to study the developed thermal turbulence with $Ra = 10^{10}$ by means of the LBGK scheme, Dixit and Babu [14] supplemented their LBM model with an interpolation procedure. Thus, non-uniform grid spacing was introduced. On the other hand, Yang et al. [15] proposed a generalized modification in the lattice Bhatnagar-Gross-Krook model. The authors added a correction term with tunable parameter in discretized lattice Boltzmann equations. This procedure removed numerical instabilities even with $Ra = 10^{12}$. However, the overall validation of the high Rayleigh number cases was not presented.

The standard multiple relaxation time (MRT) lattice Boltzmann model is very stable when considering isothermal turbulent flows. Unfortunately, this model suffers the same instabilities problems as LBGK

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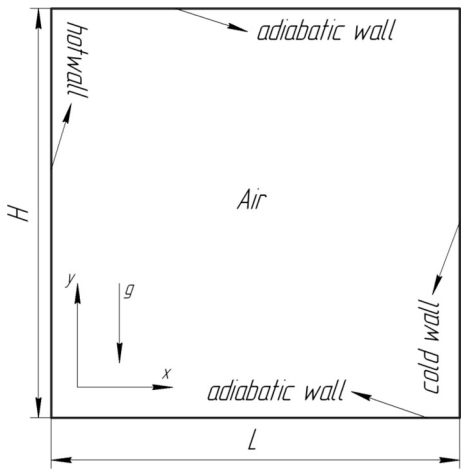


Fig. 1. Solution domain.

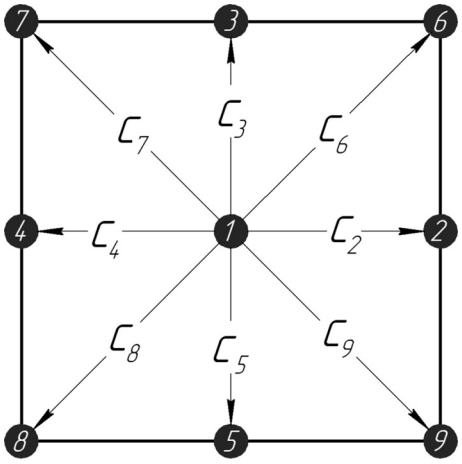


Fig. 2. D2Q9 scheme.

Table 1
Hybrid lattice Boltzmann models under study.

Collision model	BGK approximation	TRT approximation	High order regularized scheme
Convective terms approximation scheme in energy equation	The first order upwind scheme	The second order Samarsky scheme	The fourth order Runge-Kutta scheme
Accuracy/stability	Satisfactory	Average	High
Abbreviation in the present work	BGK-1	TRT-2	RLB-4

model when solving turbulent natural convection problems. On the other hand, Obrecht et al. [16] coupled the MRT model with direct numerical solution of energy equation by means of the finite difference method. It should be noted that this coupling is known as the hybrid lattice Boltzmann method [17 – 20]. Obrecht et al. performed three-dimensional simulations with the Rayleigh number up to 10^9. However, the authors used the grid size of 512^3 for stable simulations.

The high Rayleigh number flows can be accurately predicted implementing advanced collision LBM models. Sharma et al. [21] presented the Cascaded Thermal Lattice Boltzmann Method. The authors demonstrated that cascaded collision procedure led to stable simulations even with $Ra = 10^{10}$. The overall validation was performed using DNS and experimental data of other researchers. On the contrary, Frapolli et al. [22] proposed entropic multi-relaxation lattice Boltzmann model. The authors showed that this numerical scheme is stable even with $Ra = 5.17064 \cdot 10^{11}$ when analyzing the rapidly rotating Rayleigh–Bénard convection.

The above studies reflect state of the art in the field of thermal turbulence simulation via quasi-direct numerical simulation in terms of mesoscopic lattice Boltzmann method. Despite some progress in high Ra simulations, the overall validation is poorly performed when $Ra \geq 10^{10}$. Moreover, most of the studies implemented double population thermal LBM model which consumes much more random-access memory in comparison with hybrid lattice Boltzmann formulation. This issue is of great importance when implementing parallel computing on graphics processing units with limited amount of memory. Moreover, hybrid LBM models are more stable [23] and more thermodynamically consisted [24] than double population approach. The aim of this study is to introduce a novel high order hybrid lattice Boltzmann scheme for accurate prediction of thermal turbulence. Previously, the two-relaxation time (TRT) [25] and high order regularized lattice (RLB) Boltzmann [26] models coupled with the second order accurate solution of energy equation were proposed with $Ra \leq 10^{11}$. In this study, the high order regularized lattice Boltzmann scheme is combined with the energy equation solved by the fourth order Runge-Kutta scheme which extends a range of the Rayleigh number up to 10^{12}. Moreover, the cross comparison between LBGK, TRT and RLB models will be performed using the first, second and fourth order numerical schemes for energy equation solution.

2. Problem description

It is well-known that turbulence has a three-dimensional nature. However, the two-dimensional problem formulation is used in order to cover very high Rayleigh number cases. It should be stressed that this assumption will not restrict the key features of thermal turbulence in terms Tollmien–Schlichting waves formation and laminar-to-turbulent transition. The geometrical model under study is presented as a closed rectangular cavity with side heating/cooling and given in Fig. 1. It is assumed that horizontal walls are thermally-insulated. The Boussinesq approximation is applied for incompressible fluid. No volumetric heat sources are considered. Thermal radiation is negligibly small.

Fluid flow and heat transfer can be described by the mesoscopic Boltzmann equation and macroscopic energy equation as follows:

$$\frac{\partial f}{\partial t} + \mathbf{u}\nabla f = \Omega \tag{1}$$

$$\frac{\partial \Theta}{\partial \tau} + U\frac{\partial \Theta}{\partial X} + V\frac{\partial \Theta}{\partial Y} = \frac{1}{\sqrt{Pr \cdot Ra}}\left(\frac{\partial^2 \Theta}{\partial X^2} + \frac{\partial^2 \Theta}{\partial Y^2}\right) \tag{2}$$

where f is probability distribution function, t is lattice time, $\mathbf{u}$ is lattice velocity, Ω is collision operator, Θ is dimensionless temperature, τ is macroscopic time, U, V is macroscopic velocity, Pr is Prandtl number, Ra is Rayleigh number.

As was mentioned earlier, the hybrid lattice Boltzmann (HLB) formulation is used in the present work. The idea of the HLB was first proposed in [23] by Lallemand and Luo. The authors used the MRT scheme combined with the second order accurate finite difference solution of energy equation to study laminar thermal flow. On the contrary, the wide-spread and simple Bhatnagar-Gross-Krook (BGK or single relaxation time) approximation along with more advanced collision models such as the two-relaxation time and the high order regularized models will be considered in this study. Moreover, the first, the second and the fourth order of accuracy numerical schemes will be implemented when solving the energy equation by the finite difference method. Numerical diffusion is a very important property which should be carefully treated under the turbulent flow behavior. Hence, it is reasonable to couple the most numerically dissipative BGK scheme with

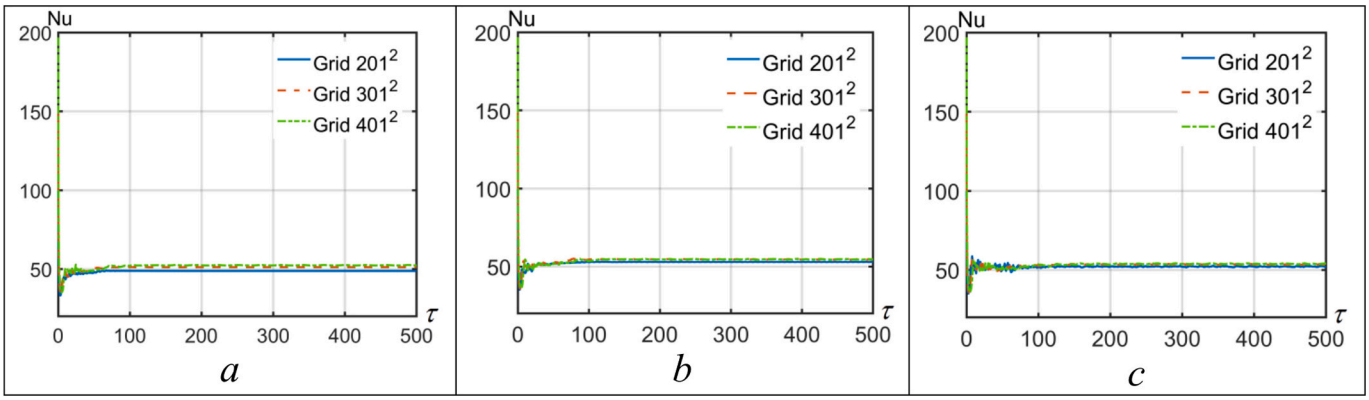


Fig. 3. Variation of mean Nusselt numbers when $Ra = 10^9$ and $Pr = 0.71$: a) BGK-1; b) TRT-2; c) RLB-4.

Table 2
Numerical performance.

Grid	BGK-1		TRT-2		RLB-4	
	CPU, s	GPU, s	CPU, s	GPU, s	CPU, s	GPU, s
501 × 126	4.15	2.81	5.21	3.37	5.73	3.73
1001 × 251	47.2	5.53	60.21	5.61	74.76	7.12
2001 × 501	240.62	14.53	305.43	16.33	347.66	20.7
4000 × 1001	1032.62	49.75	1295.27	58.84	1511.05	73.28

the first order accurate finite difference solution of energy equation. On the contrary, the most accurate high order regularized LB model is rationally to combine with the fourth order Runge-Kutta scheme for energy equation. Discretization procedure of the governing equations will be discussed below.

2.1. Bhatnagar–Gross–Krook model

The BGK approximation is simplest and numerically effective way of the Boltzmann equation solution. In accordance with this numerical scheme, the discretized equation is as follows:

$$f_k(x + c_k \cdot \Delta t_L, t_L + \Delta t_L) = f_k(x, t_L) + \Delta t_L \cdot \omega \cdot \left[f_k^{eq}(x, t_L) - f_k(x, t_L) \right] + \Delta t_L \cdot F_k \tag{3}$$

where Δt_L is lattice time step, ω is relaxation frequency, f_k^{eq} is equilibrium distribution function, F_k is force term. The subscript k gives the number of lattice Boltzmann equations in accordance with the discrete velocity set. In the present work, the two-dimensional nine-velocity (D2Q9) topology is used. Illustration of this topology is shown in Fig. 2.

The equilibrium distribution function, relaxation frequency and force term are given by:

$$f_k^{eq} = w_k \cdot \rho \cdot \left[1 + 3 \cdot (c_k \cdot \mathbf{u}) + \frac{9}{2} \cdot (c_k \cdot \mathbf{u})^2 - \frac{3}{2} \cdot (\mathbf{u} \cdot \mathbf{u}) \right] \tag{4}$$

$$\omega = \frac{1}{3 \cdot \nu + 0.5} \tag{5}$$

$$F_k = 3 \cdot \rho \cdot w_k \cdot c_y \cdot g \cdot \beta \left(T - \frac{T_h + T_c}{2} \right) \tag{6}$$

where w_k is weighting factors, ρ is density, c_k is particles speeds, ν is kinematic viscosity, g is acceleration due to gravity, β is thermal

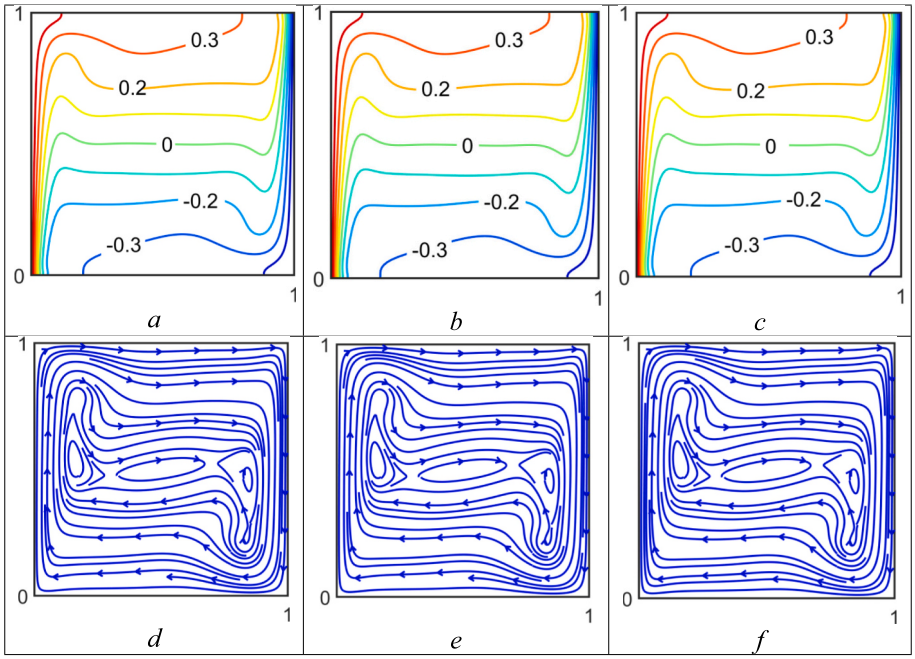


Fig. 4. Isotherms (a, b, c) and streamslices (d, e, f) when $Ra = 10^6$ and $Pr = 0.71$: a, d) BGK-1; b, e) TRT-2; c, f) RLB-4.

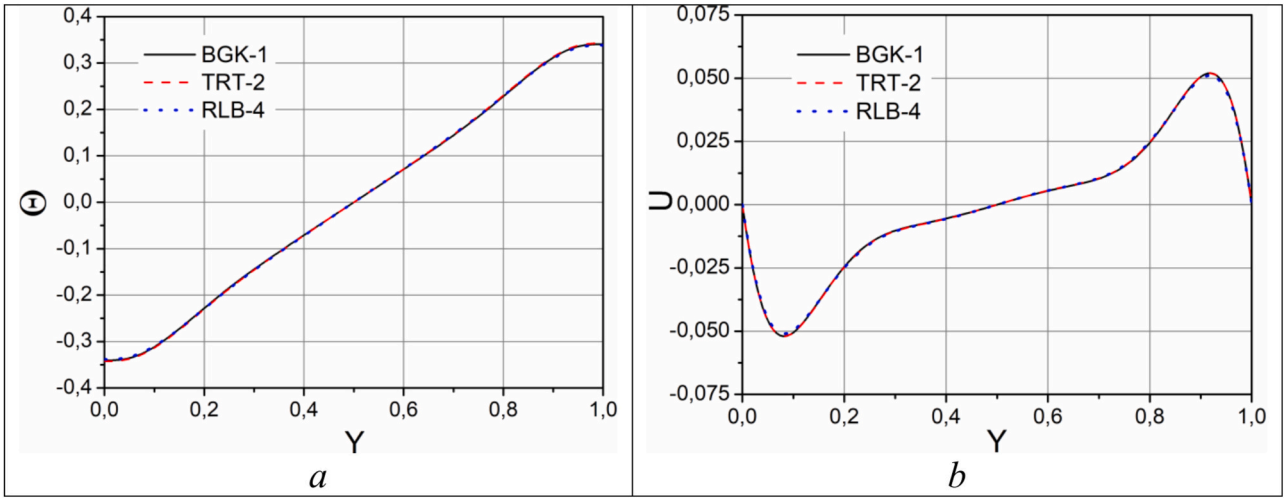


Fig. 5. Variations of temperature (a) and horizontal velocity component (b) when $Ra = 10^6$ and $Pr = 0.71$.

Table 3
Mean Nusselt numbers at the hot wall.

Ra	[10]	[14]	[31,32]	[33]	BGK-1	TRT-2	RLB-4
10^4	2.25	2.26	2.25	–	2.21	2.21	2.2
10^5	4.52	4.52	4.52	–	4.43	4.44	4.42
10^6	8.83	8.51	8.83	–	8.71	8.77	8.71
10^7	16.52	16.79	16.52	16.52	16.3	16.48	16.37
10^8	30.22	30.51	30.23	30.23	29.48	30.06	29.72

expansion.

Weighting factors (w_k) and particle speeds (c_k) are defined as follows:

$$w_k = \begin{cases} \frac{4}{9}; k = 1 \\ \frac{1}{9}; k = 2..5 \\ \frac{1}{36}; k = 6..9 \end{cases}$$

$$c_k = \begin{cases} (0,0); k = 1 \\ c([cos(i-1)\pi/2\,], sin[(i-5)\pi/2+\pi/4\,]); k = 2..4 \\ c\sqrt{2}\,(cos[(i-5)\pi/2+\pi/4\,], sin[(i-5)\pi/2+\pi/4\,]); k = 5..9 \end{cases}$$

Within lattice Boltzmann method, macroscopic variables are recovered as

$$\rho(x,t) = \sum_{k=1}^{9} f_k(x,t). \tag{7}$$

$$\mathbf{u}(x,t) = \frac{1}{\rho}\cdot\sum_{k=1}^{9} c_k\cdot f_k(x,t). \tag{8}$$

2.2. Two relaxation time model

Generally speaking, the TRT scheme [27] is a special form of the multiple relaxation time model. The two relaxation time model simplifies the MRT scheme adding only two distinct relaxation frequencies. One relaxation frequency stands for the symmetric part of the distribution functions whereas another one covers the asymmetric part. The discretized form of the lattice Boltzmann equations using TRT scheme is as follows:

$$f_k\left(\vec{x}+\vec{c}_k\cdot\Delta t_L, t_L+\Delta t_L\right) = \\ = f_k(\vec{x},t) - \Delta t_L\cdot\omega_s\cdot\left(f_k^s - f_k^{seq}\right) - \Delta t_L\cdot\omega_a\cdot\left(f_k^a - f_k^{aeq}\right) + \Delta t_L\cdot F_k \tag{9}$$

$$f_k^s = \frac{f_k + f_{-k}}{2}, f_k^{seq} = \frac{f_k^{eq} + f_{-k}^{eq}}{2}, \\ f_k^a = \frac{f_k - f_{-k}}{2}, f_k^{aeq} = \frac{f_k^{eq} - f_{-k}^{eq}}{2}. \tag{10}$$

$$\omega_s = \frac{1}{3\cdot\nu + 0.5} \tag{11}$$

$$\left(\frac{1}{\omega_s} - \frac{1}{2}\right)\cdot\left(\frac{1}{\omega_a} - \frac{1}{2}\right) = \frac{1}{4} \tag{12}$$

where f_k^s is symmetrical part of distribution function, f_k^{seq} is symmetrical part of equilibrium distribution function, f_k^a is asymmetrical part of distribution function, f_k^{aeq} is asymmetrical part of equilibrium distribution function, ω_s is symmetrical relaxation frequency, ω_a is asymmetrical relaxation frequency.

Note that the same two-dimensional nine-velocity is used. Hence, values of weighting factors and particle speeds are the same as in the case of the BGK model.

2.3. High order regularized lattice Boltzmann model

Numerical stability of the LBM model can be significantly enhanced using regularization techniques and higher-order corrections to the collision operator. In particular, high order regularized lattice Boltzmann model proposed in [28] is very stable and it accurately captures turbulent patterns. The governing equations in discretized form are as follows:

$$f_k(x + c_k\cdot\Delta t_L,\ t+\Delta t_L) = \\ = f_k^{eq}(x,\ t) + \Delta t_L\cdot(1-\omega)\cdot f_k^{neq}(x,\ t) + \Delta t_L\cdot F_k \tag{13}$$

$$f_k^{eq} = w_k\cdot\rho\cdot\left[\begin{array}{c} 1 + \frac{u_\alpha\cdot c_{k,\alpha}}{c_s^2} + \frac{u_\alpha\cdot u_\beta}{2\cdot c_s^4}\left(c_{k,\alpha}\cdot c_{k,\beta} - c_s^2\cdot\delta_{\alpha\beta}\right) + \\ + \frac{u_x^2\cdot u_y\cdot e_{k,y}}{2\cdot c_s^6}\left(c_{k,x}^2 - c_s^2\right) + \frac{u_y^2\cdot u_x\cdot c_{k,x}}{2\cdot c_s^6}\times \\ \times\left(c_{k,y}^2 - c_s^2\right) + \frac{u_x^2\cdot u_y^2}{4\cdot c_s^8}\left(c_{k,x}^2 - c_s^2\right)\cdot\left(c_{k,y}^2 - c_s^2\right) \end{array}\right] \tag{14}$$

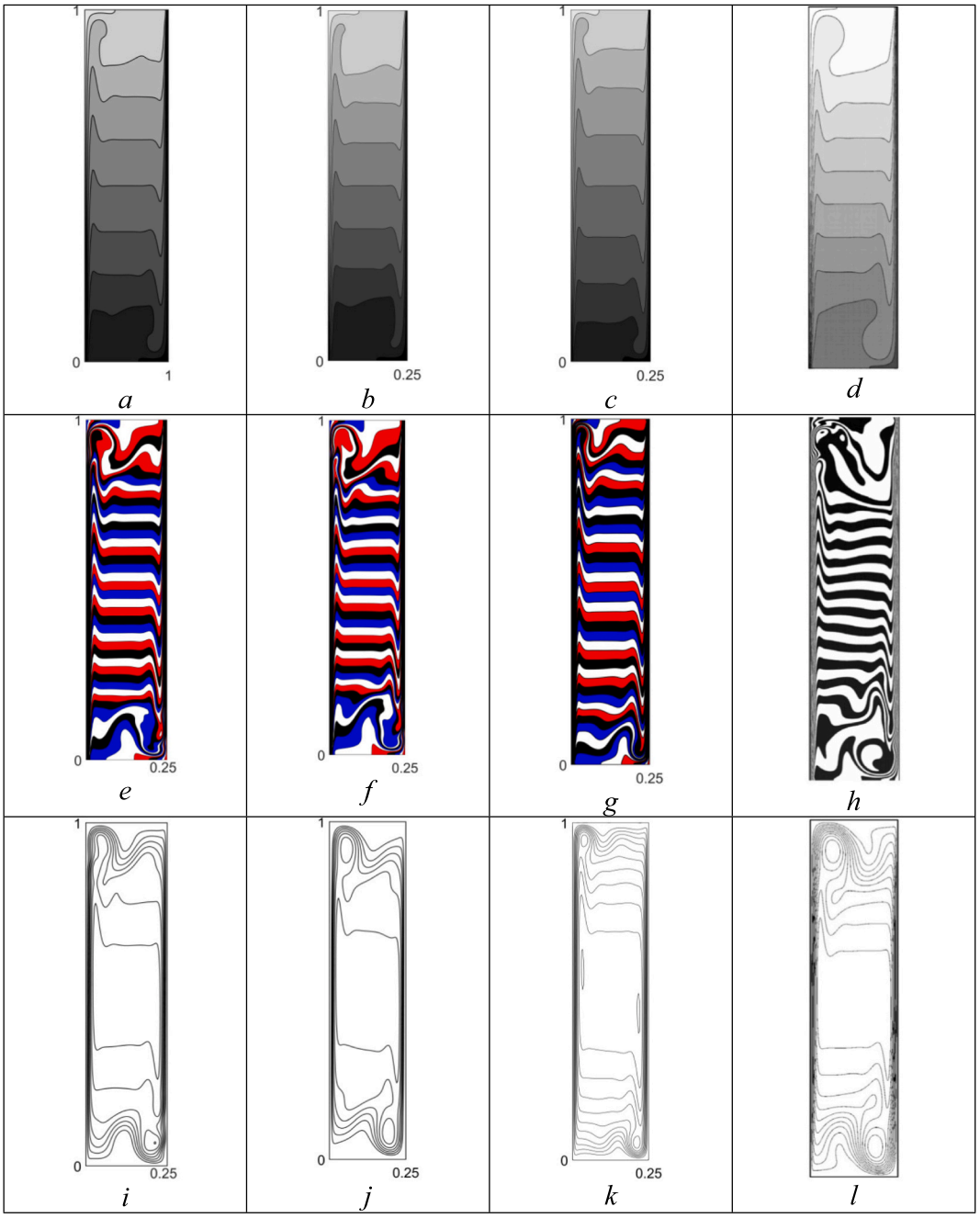


Fig. 6. Time-averaged (a, b, c, d) and instantaneous (e, f, g, h) thermal fields, time-averaged streamlines (i, j, k, l) when $Ra = 6.4{\cdot}10^8$ and $Pr = 0.71$: a, e, i) BGK-1; b, f, g) TRT-2; c, g, k) RLB-4; d, h, l) Trias et al. [34].

$$f_k^{neq} = w_k \cdot \left[\begin{array}{c} \frac{\tau_{\alpha\beta}}{2 \cdot c_s^4} \left(c_{k,\alpha} \cdot e_{k,\beta} - c_s^2 \cdot \delta_{\alpha\beta} \right) + \\ + \frac{\left(u_y \cdot \xi_{xx} + 2 \cdot u_x \cdot \xi_{xy} \right) \cdot c_{k,y}}{2 \cdot c_s^6} \left(c_{k,x}^2 - c_s^2 \right) + \\ + \frac{\left(u_x \cdot \xi_{yy} + 2 \cdot u_y \cdot \xi_{xy} \right) \cdot c_{k,x}}{2 \cdot c_s^6} \left(c_{k,y}^2 - c_s^2 \right) + \\ + \frac{\left(u_y^2 \cdot \xi_{xx} + 4 \cdot u_x \cdot u_y \cdot \xi_{xy} + u_x^2 \cdot \xi_{yy} \right)}{4 \cdot c_s^8} \left(c_{k,x}^2 - c_s^2 \right) \times \\ \times \left(c_{k,y}^2 - c_s^2 \right) \end{array} \right] \quad (15)$$

$$\xi_{\alpha\beta} = \sum_{k=1}^{9} c_{k,\alpha} \cdot c_{k,\beta} \cdot \left(f_k(x,t) - f_k^{eq}(x,t) \right), \alpha = x,y; \beta = x,y \quad (16)$$

where f_k^{neq} is non-equilibrium distribution function, $\tau_{\alpha\beta}$- viscous stress tensor, δ - Kronecker delta.

Note that the same D2Q9 scheme is implemented as in the cases of BGK and TRT models.

2.4. Algorithm description

Numerical procedure within the hybrid lattice Boltzmann method is very similar to the double-population thermal LB model. The mesoscopic lattice Boltzmann equations are solved via the particles collision –

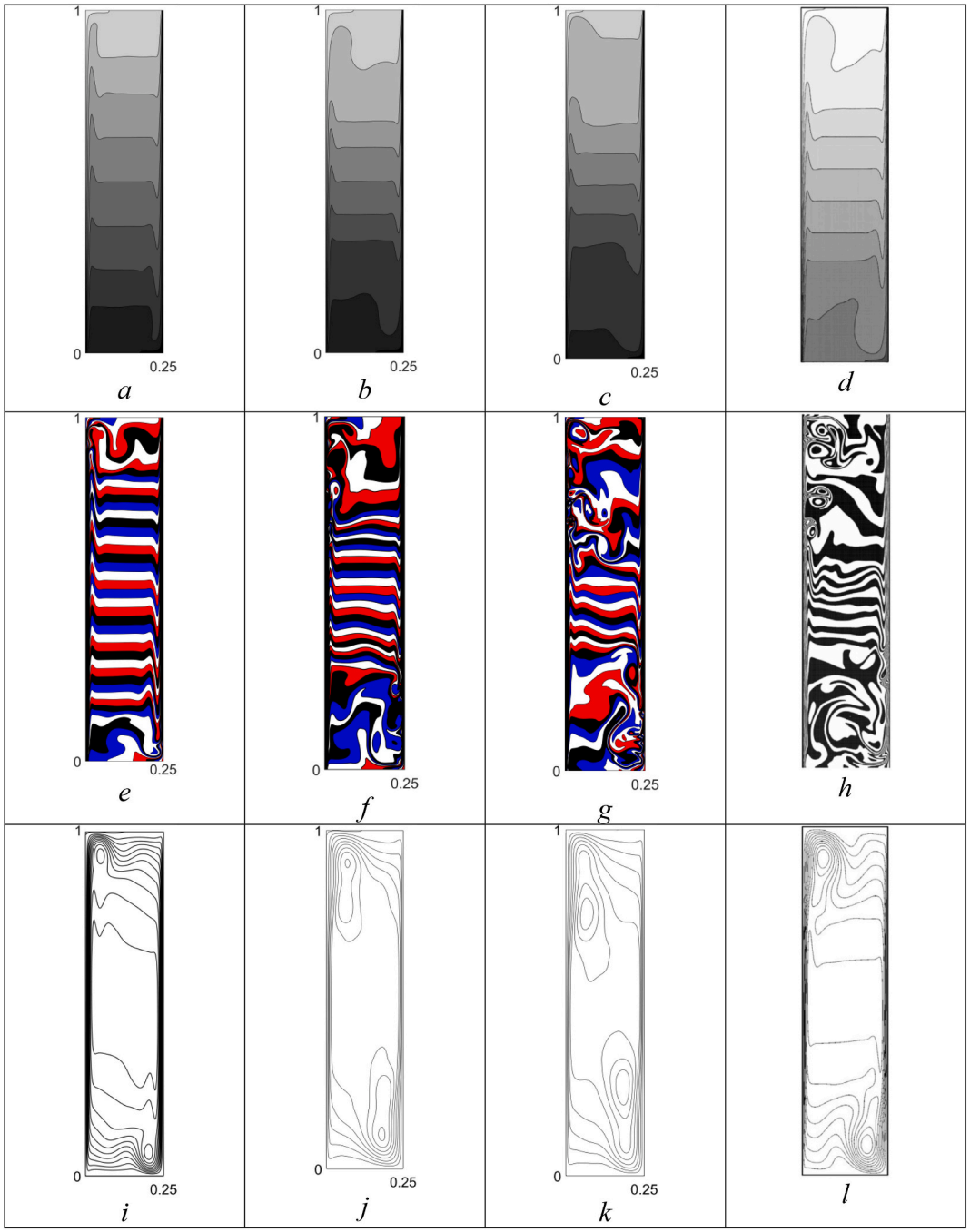


Fig. 7. Time-averaged (a, b, c, d) and instantaneous (e, f, g, h) thermal fields, time-averaged streamlines (i, j, k, l) when $Ra = 2{\cdot}10^9$ and $Pr = 0.71$: a, e, i) BGK-1; b, f, g) TRT-2; c, g, k) RLB-4; d, h, l) Trias et al. [34].

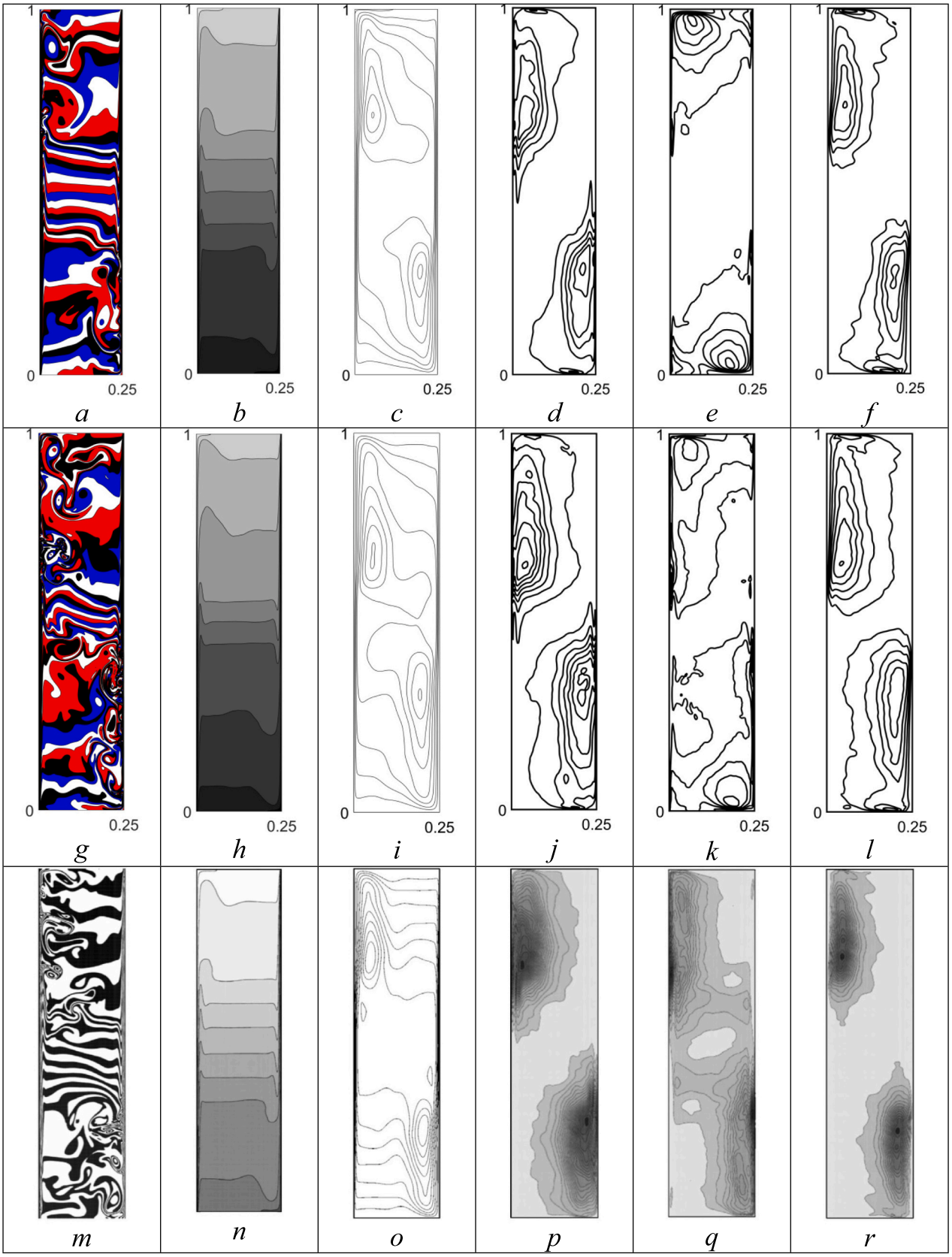


Fig. 8. Instantaneous (a, g, m) and time-averaged (b, h, n) thermal fields, time-averaged streamlines (c, i, o), turbulent kinetic energy (d, j, p), temperature variance (e, k, q) and root mean square horizontal velocity (f, l, r) when $Ra = 10^{10}$ and $Pr = 0.71$: a, b, c, d, e, f) TRT-2; g, h, i, j, k, l) RLB-4; m, n, o, p, q, r) Trias et al. [34].

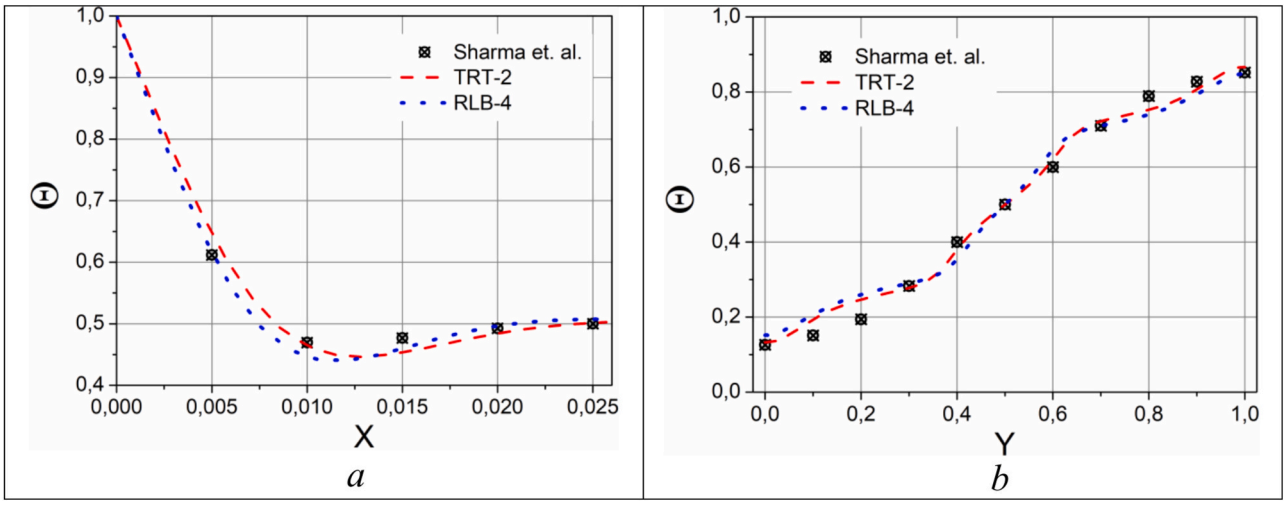


Fig. 9. Temperature variations when $Ra = 10^{10}$ in the sections of: a) Y = 0.5; b) X = 0.5.

propagation algorithm. After that, the energy equation is solved by the finite difference method instead of additional collision – propagation algorithm for temperature calculation. In the case of the BGK scheme and the first order accurate finite difference solution of energy equation, the numerical procedure is as follows:

- Collision step:

$$f_k(x + c_k \cdot \Delta t_L, t_L + \Delta t_L) = f_k(x, t_L) + \Delta t_L \cdot \omega \cdot \left[f_k^{eq}(x, t_L) - f_k(x, t_L) \right] + \Delta t_L \cdot F_k$$

- Propagation step:

$$f_k(x + c_k \cdot \Delta t_L, t_L + \Delta t_L) = f_k(x, t_L)$$

- Energy step:

$$T_{ij}^{n+1} = T_{ij}^{n} + \frac{1}{\sqrt{Pr \cdot Ra}} \cdot \Delta\tau \cdot \left(\frac{T_{i+1j}^{n} - 2 \cdot T_{ij}^{n} + T_{i-1j}^{n}}{h_x^2} + \frac{T_{ij+1}^{n} - 2 \cdot T_{ij}^{n} + T_{ij-1}^{n}}{h_y^2} \right) -$$

$$0.5 \cdot \Delta\tau \cdot \left(U_{ij}^{n} + \left| U_{ij}^{n} \right| \right) \cdot \left(\frac{T_{ij}^{n} - T_{i-1j}^{n}}{h_x} \right) - 0.5 \cdot \Delta\tau \cdot \left(U_{ij}^{n} - \left| U_{ij}^{n} \right| \right) \cdot \left(\frac{T_{i+1j}^{n} - T_{ij}^{n}}{h_x} \right) -$$

$$0.5 \cdot \Delta\tau \cdot \left(V_{ij}^{n} + \left| V_{ij}^{n} \right| \right) \cdot \left(\frac{T_{ij}^{n} - T_{ij-1}^{n}}{h_x} \right) - 0.5 \cdot \Delta\tau \cdot \left(V_{ij}^{n} - \left| V_{ij}^{n} \right| \right) \cdot \left(\frac{T_{ij+1}^{n} - T_{ij}^{n}}{h_x} \right).$$

It should be stressed that the attention was given to discretization of convective terms since these terms have the highest impact on numerical viscosity. Hence, the LBGK model is coupled with the first order approximation of energy equation using the first upwind scheme whereas the monotonic Samarsky scheme is combined with the TRT model. On the contrary, the high order regularized lattice Boltzmann scheme is incorporated with the fourth order Runge-Kutta method for energy equation. For better understanding, the models under consideration are given in Table 1.

It should be noted that a single thread central processing unit (CPU) computing is very inefficient when using direct numerical simulation method. In this regard, an in-house MatLab code was developed with parallel algorithm implementation on graphics processing units (GPU). Comparison of numerical performance between CPU and GPU codes will be discussed in the next sub-section.

2.5. Grid study and performance

When using uniform grid spacing, it is computationally ineffective to resolve the finest Kolmogorov scales near the solid walls. In order to overcome this weakness, a pseudo-direct numerical simulation (pDNS) is recommended. It is a CFD technique for turbulence simulation [29,30]. This technique is quite similar to classical direct numerical simulation method. However, the key distinction between the DNS and pDNS lies in grid resolution. The direct numerical simulation approach assumes full resolution of Kolmogorov spatial and temporal scales. On the contrary, the finest scales can be ignored when using the pseudo-DNS method. In general, the idea of the pseudo-direct numerical simulation technique is close to the large eddy simulation (LES) where the smallest length scales are ignored. But unlike the LES, the pDNS is turbulence model free. Hence, there is no need in any semi-empirical data to initialize simulations. Moreover, the lattice Boltzmann method operates on a mesoscopic level where there are no Kolmogorov scales of time and space. Thus, the hybrid lattice Boltzmann models proposed in this study are considered as the pDNS tool. The grid sensitivity test is performed in terms of Nusselt numbers as functions of dimensionless time. Fig. 3 presents the dynamics of mean Nusselt numbers.

As could be expected, the order of approximation affects the number of grid points necessary for solution convergence. The higher accuracy of the scheme is the lower mesh density is needed. The BGK-1 scheme is barely converged with 401^2 grid points whereas high order regularized lattice Boltzmann model coupled with the fourth order Runge-Kutta scheme for energy equation provides mesh-insensitivity solution with grid size of 201^2. It should be noted that the Rayleigh number significantly affects the number of mesh points needed for grid-independent solution. Hence, the grid study was performed every time when the *Ra* was increased.

When performing high fidelity simulations, the code execution time is a crucial parameter which affects practicability of computations. For example, conventional direct numerical simulation techniques consume a huge amount of computing resources. Nevertheless, the execution times are far away from the implementation of the DNS in real engineering systems. In the present work, numerical performance of the hybrid lattice Boltzmann models under study is examined in terms of the CPU and GPU times. The Intel Core i5–13,400 was used as the central processing unit in a single thread computing mode whereas the parallel computations were performed on the NVIDIA GTX 1060 6 GB. The results are presented in Table 2. Note that the CPU and GPU times are given for 1000 iterations.

As could be predicted, the BGK collision model along with the first order accurate finite difference solution of the energy equation is the

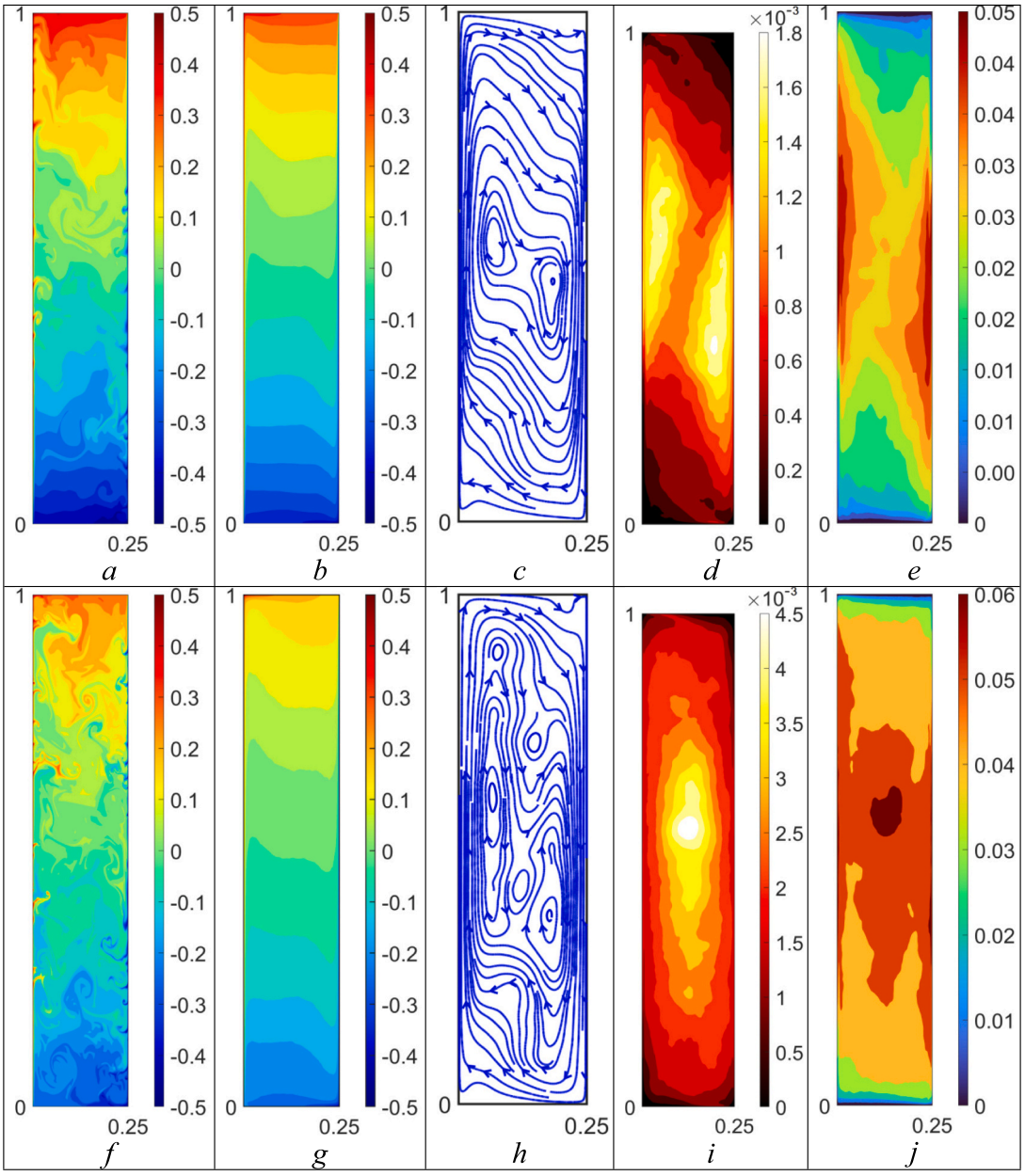


Fig. 10. Instantaneous (a, f) and time-averaged (b, g) thermal fields, time-averaged streamlines (c, h), turbulent kinetic energy (d, i), and root mean square vertical velocity (e, j) when $Ra = 10^{11}$ and $Pr = 0.71$: a, b, c, d, e) TRT-2; f, g, h, i, j) RLB-4.

most numerically efficient. However, the difference in computational speed between BGK-1, TRT-2 and RLB-4 is around 20 % and 32 %, respectively, when using the CPU computing. The same pattern is remained unchanged with the use of GPU at high grid sizes. The most numerically intensive model is the RLB-4 due to additional computations during the energy equation and collision steps. The TRT-2 model outperforms the RLB-4 model by a factor of 1.25. Hence, a reasonable recommendation between accuracy and computational performance will be provided in the present work in accordance with the Rayleigh number.

3. Results and discussion

Numerical experiments are focused on the cross comparison of the results obtained by different models and schemes. A huge range of the Rayleigh number of $10^4 \leq Ra \leq 10^{12}$ is covered. Both the first and the second order statistics are analyzed in detail. Numerical results are given in terms of instantaneous and time-averaged temperature and streamline contours, turbulent kinetic energy, temperature variance, root mean square velocity and mean Nusselt numbers. Three sub-sections are presented depending on the Rayleigh number range. In general, laminar, transitional and fully-developed turbulent flows are discussed.

3.1. *Laminar flow* $10^4 \leq Ra \leq 10^8$

Thermal and flow fields computed by different models are given in Fig. 4. In order to obtain a steady-state solution, convergence criteria were set 10^{-7}.

It is interesting to note that all hybrid models under study reproduce fully identical results in terms of both quantitative and qualitative data. Probably, this behavior is associated with fully laminar flow without any significant bifurcation where the first order approximation is enough to capture all relevant scales. In this regard, the Rayleigh number was increased up to 10^7. Temperature and velocity profiles are presented in

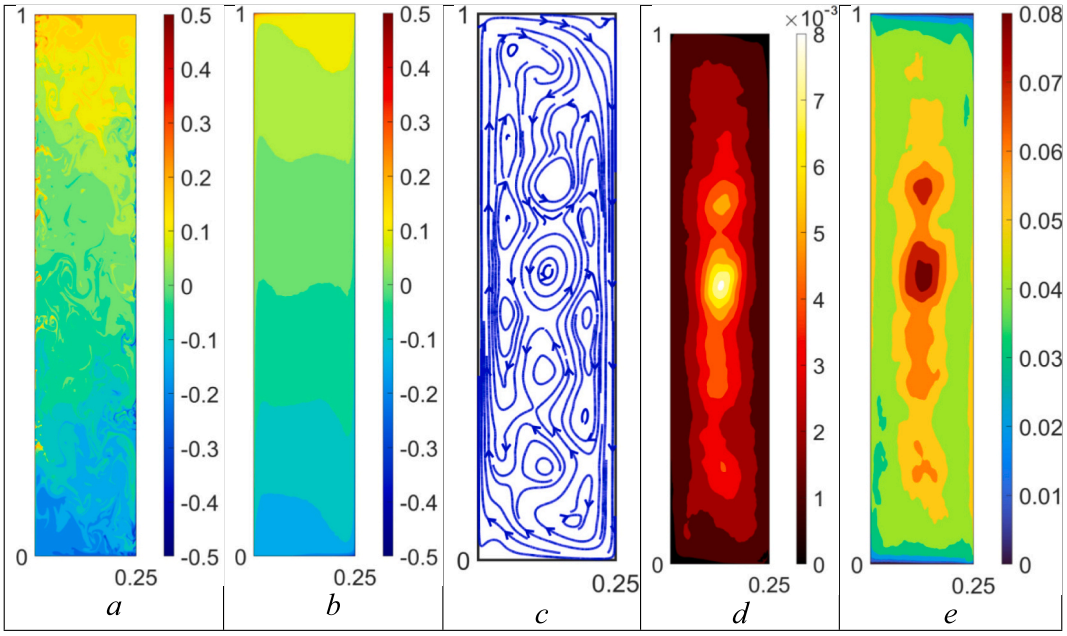


Fig. 11. Instantaneous (a) and time-averaged (b) thermal fields, time-averaged streamlines (c), turbulent kinetic energy (d), and root mean square vertical velocity (e) when $Ra = 10^{12}$ and $Pr = 0.71$.

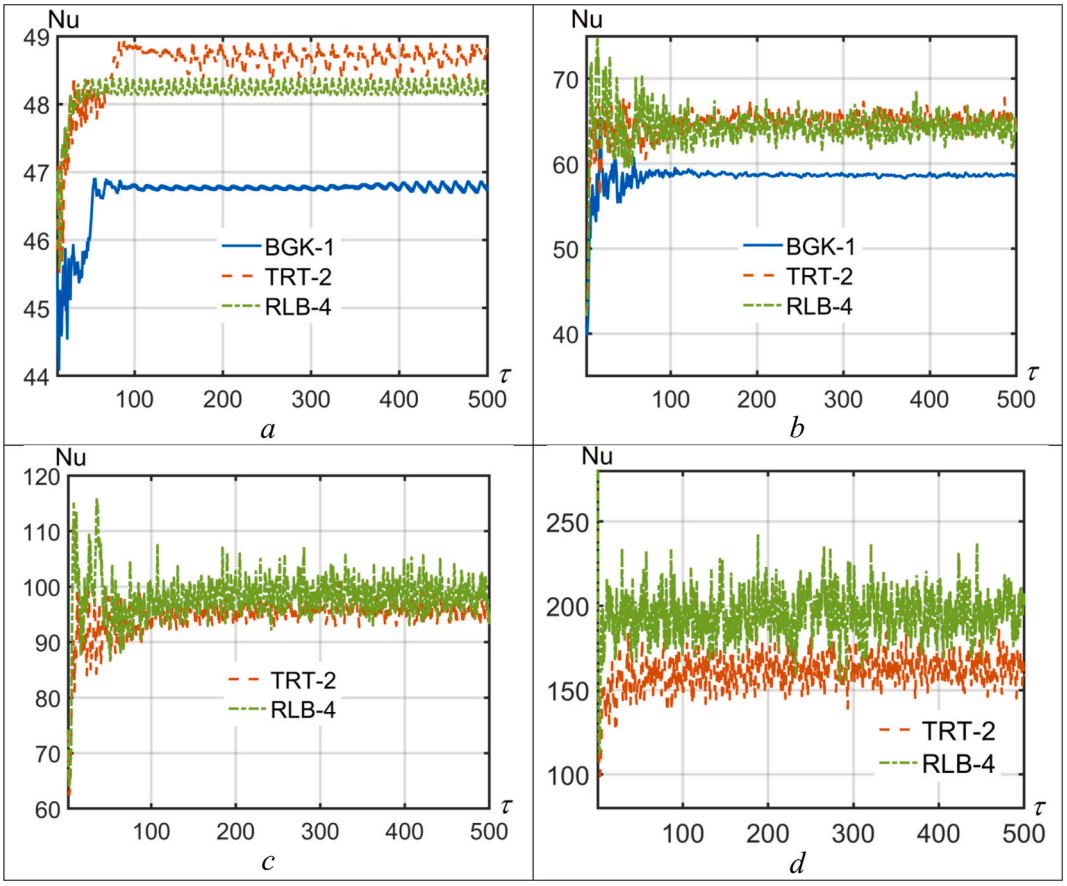


Fig. 12. Mean Nusselt numbers dynamics: a) $Ra = 6.4{\cdot}10^8$; b) $Ra = 2{\cdot}10^9$; c) $Ra = 10^{10}$; d) $Ra = 10^{11}$.

Fig. 5.

It is clearly seen that the BGK-1, TRT-2 and RLB-4 models predict the same values of temperature and velocity. No significant deviations are observed under laminar flow behavior in terms of local heat transfer and fluid flow characteristics. Integral data are given in Table 3.

When analyzing the data presented in Table 3, it is found that the mean Nusselt numbers are in a very good agreement with numerical results of other researchers. The error in integral characteristics does not

Table 4
Time-averaged mean Nusselt numbers.

Ra	2D [36]	2D [34]	2D [21]	3D [2]*	3D [37,38]	3D [35]	BGK-1	TRT-2	RLB-4
$6.4{\cdot}10^8$	49.2	49.23	–	49.61	45.16	49.24	47.55	48.91	48.09
$2{\cdot}10^9$	66.5	66.4	–	68.4	59.44	66.63	60.16	64.79	63.74
10^{10}	101	100.6	100.2	107.69	94.21	101.94	–	94.74	98.11
10^{11}	–	–	–	–	154.6	194.48	–	160.44	194.87
10^{12}	–	–	–	–	–	–	–	–	391.55

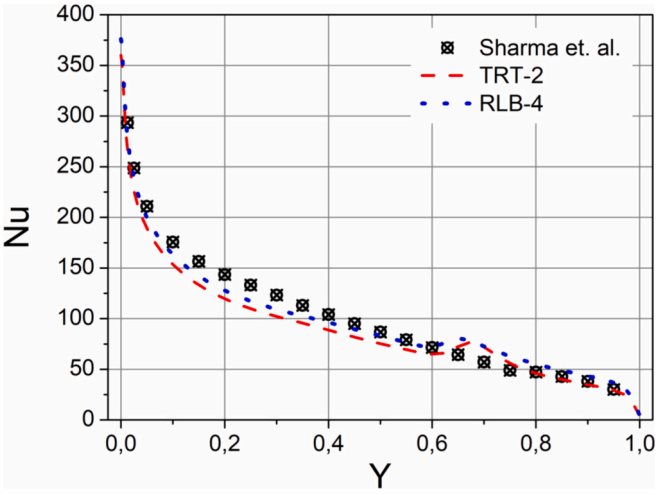


Fig. 13. Local Nusselt numbers when $Ra = 10^{10}$.

exceed 3 %. It is interesting to note that BGK-1 and RLB-4 reproduce almost the same results since these models are single relaxation time approximation based. On the other hand, the most accurate model is TRT-2 in the laminar case under consideration. Probably, two relaxation times dealing with both energy and viscosity fluxes better resolve near wall regions. In general, all hybrid lattice Boltzmann models under study can successfully deal with natural convection in a differentially heated cavity in a range of $10^4 \leq Ra \leq 10^8$.

3.2. *Laminar-to-turbulent transition* $6.4{\cdot}10^8 \leq Ra \leq 10^{10}$

The overall assessment of hybrid lattice models under consideration is performed using benchmark DNS data of Trias et al. [34]. Instantaneous and time-averaged temperature contours, streamlines, turbulent kinetic energy, temperature variance and root mean square velocity (U_{rms} and V_{rms}) will consecutively cross compared when increasing the Rayleigh number. In order to do a quantitative analysis, the mean Nusselt numbers will be considered. The data under study are given in Fig. 6.

When $Ra = 6.4{\cdot}10^8$, the buoyancy driven flow becomes unsteady due to irregular thermal behavior at the upstream and downstream corners. The rest of the cavity is uniformly heat stratified. Unexpectedly, the less accurate model in terms of outer flows prediction is RLB-4. However, this model accurately reproduces streamlines configuration in the core of the cavity unlike BGK-1 and TRT-2 schemes. The discrepancy can be explained by the numerical diffusion. The most accurate RLB-4 model smoothed flow fluctuations near the horizontal boundaries with this value of the Rayleigh number. It is found that BGK-1 and TRT-2 schemes predict very similar instantaneous and time-averaged thermal fields. On the contrary, a slight difference in flow pattern is observed. Temperature and streamline contours illustrating low level thermal buoyancy turbulence are given in Fig. 7.

As could be predicted, the worst performance is found when using the BGK-1 model to simulate low level thermal turbulence. This model is unable to catch and to resolve turbulent features of the buoyancy driven flow due to a slightly lower value of relaxation frequency set for numerical stability. On the contrary, the TRT-2 scheme successfully reproduces a chaotic behavior near the adiabatic walls. Along with that, this numerical scheme overpredicts the level of heat stratification. However, the time-averaged characteristics are in a satisfactory agreement with the benchmark data of Trias et al. [34]. The RLB-4 model gives the best results in terms of the instantaneous thermal field. The core of the solution domain remains thermally stratified whereas the flow oscillates at the rest of the cavity. It is interesting to note that the models under consideration provide almost the same streamline contours which are in a satisfactory agreement with reference data [34]. It should be noted that further increase in the buoyancy force makes BGK-1 scheme unstable. Hence, numerical results computed by the TRT-2 and RLB-4 schemes will be presented hereinafter. The first and the second order statistics are given in Fig. 8.

When comparing numerical results with reference data, it might be seem that the TRT-2 model outperforms the RLB-4. Indeed, the first order statistics are in a very good agreement with Navier-Stokes results of Trias et al. [34]. However, the TRT-2 scheme underpredicts the second order statistics. Especially, it is clearly seen when analyzing the temperature variance. In general, all models under consideration reproduce similar results close to benchmark DNS data. Strong buoyancy force destratifies almost the entire solution domain. Heat stratification is observed only in a small region of the core of the cavity. The TRT-2 model slightly overstates the level of thermal stratification which indicates turbulence underprediction. On the contrary, the RLB-4 slightly overestimates the first order statistics. Along with that, a discrepancy in the temperature variance is observed at the bottom left and top right corners of the cavity. On the other hand, turbulent kinetic energy and root mean square velocity are much better agreed with reference data than those obtained by the TRT-2 scheme.

Additional comparative study of time-averaged temperature profiles computed for the case of a square cavity as in [21] is presented in Fig. 9.

When analyzing the data given in Fig. 8, it is found that both the TRT-2 and RLB-4 models reproduce the same temperature trend as in the case of the cascaded (CLBM) multiple relaxation time lattice Boltzmann scheme [21] in the cross section of $Y = 0.5$. A slightly better agreement is observed when using the RLB-4 model. On the other hand, a very good mapping of temperatures is found at the core of the cavity in the cross section of $X = 0.5$. It is interesting to note that temperature values are well matched at the horizontal boundaries whereas the deviation is revealed at the near-wall regions. However, this deviation is not crucial and, probably, this behavior is associated with different grid sizes. Along with that, the disagreement might be affected by the different number of iteration in order to achieve the statistically steady-state solution.

3.3. *Developed turbulence* $10^{11} \leq Ra \leq 10^{12}$

The first and the second order statistics computed by the TRT-2 and RLB-4 models for developed thermal turbulence are given in Fig. 10.

As in the previous case with $Ra = 10^{10}$, the TRT-2 scheme predicts turbulent characteristics with lower intensity. When comparing thermal fields, it is found that similar temperature values are observed in the core of the cavity. On the contrary, the thermal behavior is significantly different near the bottom and top walls. The RLB-4 model reproduces higher fluid turbulent mixing due to better resolution of low-scale

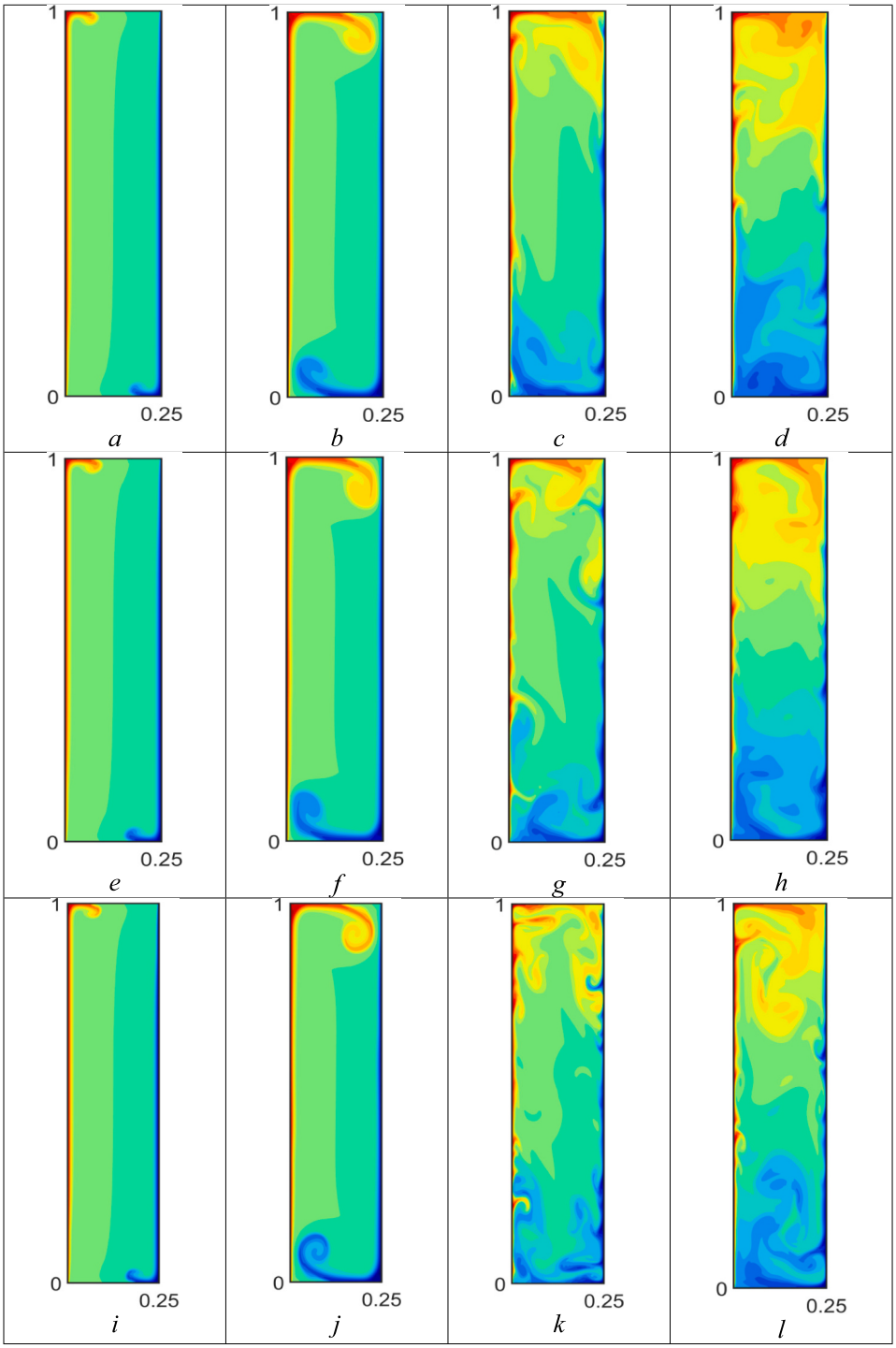


Fig. 14. Thermal fields computed by BGK-1 (a, b, c, d), TRT-2 (e, f, g, h) and RLB-4 (i, j, k, l) when $Ra = 2{\cdot}10^9$: a, e,i) $\tau = 2$; b, f,j) $\tau = 4$; c, g,k) $\tau = 10$; d, h, l) $\tau = 20$.

structures. Thermal plumes are generated on the most part of isothermal boundaries which reduces/increases temperature upstream/downstream areas. Moreover, the RLB-4 scheme predicts more complex flow pattern even under the time-averaging procedure. In particular, several convective cells are formed in the core of the cavity. Despite a general similarity of the first order statistics computed by TRT-2 and RLB-4 models, the second order statistics are significantly differed. The TRT-2 scheme still predicts laminar regions at the bottom left and top right corners of the solution domain. On the other hand, turbulent kinetic energy is distributed in the entire cavity when using the RLB-4 model. Along with that, the vertical component of the velocity vector fluctuates more intensively. It should be stressed that the turbulent flow is fully developed with $Ra = 10^{11}$. Nevertheless, laminar regions in terms of turbulent kinetic energy distribution still exists at the bottom left and top right corners of the cavity. It is found that the TRT-2 model underestimates turbulence intensity. On the contrary, the RLB-4scheme successfully resolves turbulent structures due to high order approximation. It is very important to stress that the RLB-4 model is stable even with $Ra = 10^{12}$. The first and second order statistics are given in Fig. 11.

As could be expected, flow pattern becomes more complex and

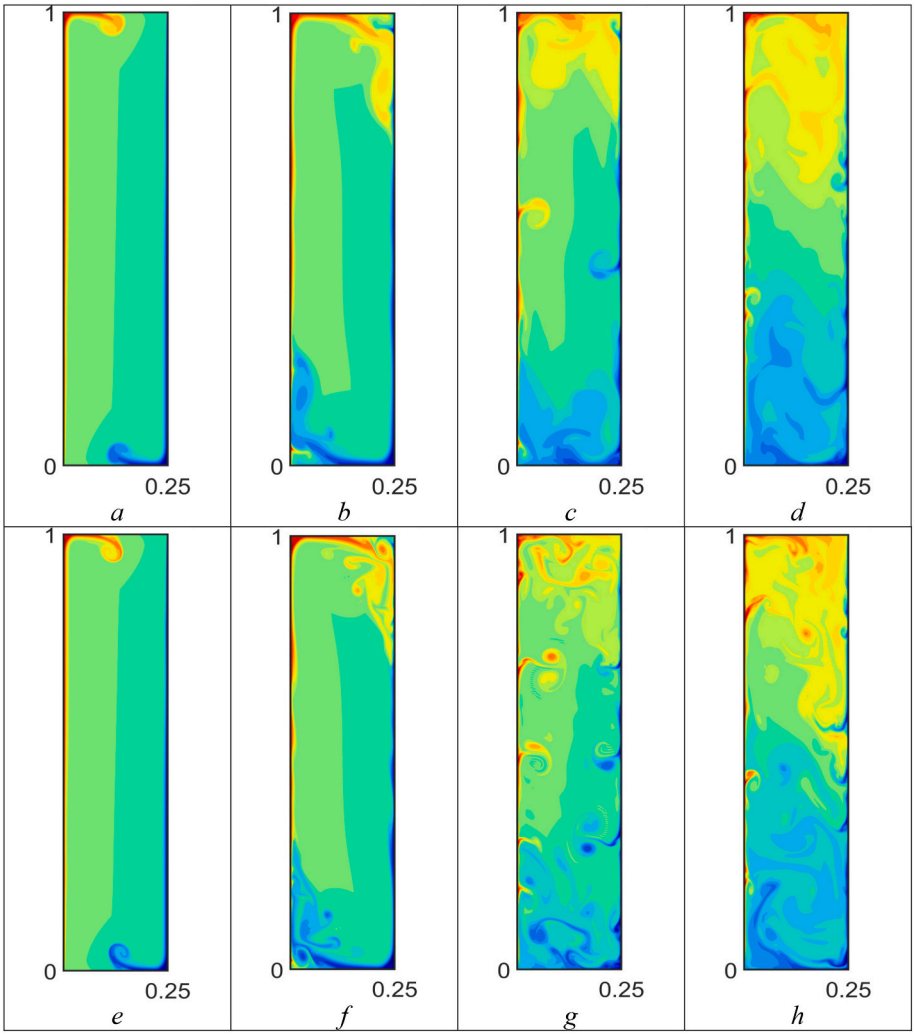


Fig. 15. Thermal fields computed by TRT-2 (a, b, c, d) and RLB-(e, f, g, h) when $Ra = 10^{10}$: a, e) $\tau = 3$; b, f) $\tau = 6$; c, g) $\tau = 12$; d, h) $\tau = 24$.

chaotic even under time-averaging procedure with further increment in the buoyancy force. It is found that temperature at the upstream/downstream regions is decreased/increased. Along with that, turbulent kinetic energy concentrates at the core of the cavity. It is interesting to note that low-scale structures such as thermal plumes generated along isothermal walls have no effect on TKE as in the case of $Ra \leq 10^{10}$. Hence, turbulent kinetic energy is predominantly transferred by large eddies formed at the core of the solution domain. Moreover, oscillation of vertical component of velocity vector is also shifted from the isothermal walls to the center of the cavity. It is well-knows that an increase in the Rayleigh number leads to a decrease in the boundary layer thickness and heat destratification of the thermal field. These phenomena were successfully reproduced by the TRT-2 and RLB-4 models. However, a shift in the second order statistics from upstream and downstream corners to the core of the cavity was revealed by means of the RLB-4 scheme.

3.4. Heat transfer rate

Variations of mean Nusselt numbers with time are presented in Fig. 12.

When analyzing the data given in Fig. 10, it is found that the BGK-1 underestimates the heat transfer rate in comparison with TRT-2 and RLB-4 schemes. On the other hand, the TRT-2 model predicts slightly higher values of mean Nusselt numbers than RLB-4 model with $Ra \leq 2{\cdot}10^9$. Probably, this pattern is concerned with higher numerical dissipation of TRT-2 scheme. When $Ra \geq 10^{10}$, the opposite behavior is observed. Moreover, the magnitude of oscillations predicted by the TRT-2 model is significantly lower in comparison with RLB-4 model. The time-averaged data are given in Table 4.

As could be assumed, the BGK-1 and TRT-2 models underpredicts heat transfer rate when $Ra \geq 2{\cdot}10^9$ and $Ra \geq 10^{10}$, respectively. On the contrary, the RLB-4 scheme successfully reproduces mean Nusselt numbers even without grid refinement procedure. However, the TRT-2 model satisfactory predicts Nu values along with the first and second order statistics when $Ra \leq 10^{10}$. On the other hand, it gives an error of 17.5 % in the mean Nusselt number. The RLB-4 scheme successfully predicts integral characteristics of developed thermal turbulence.

Variation of local Nusselt numbers along the hot wall of the square cavity is given in Fig. 13.

It is found that the RLB-4 model has a slightly better agreement with the data computed by the cascaded multiple relaxation time lattice Boltzmann scheme [21] than the TRT-2 approach. However, the models developed in this study and the CLBM of Sharma et al. [21] differently predict the point where the laminar boundary layer transits to the turbulent one. Unfortunately, temperature contours are unavailable in the reference work. Hence, it is troublesome to find out the reason of this deviation. Probably, this mismatch might be concerned with the different thermal LB models used in the present work and in [21]. It should be stressed that the local Nusselt numbers are suddenly increased at the curve due to thermal plumes generation produced by turbulence.

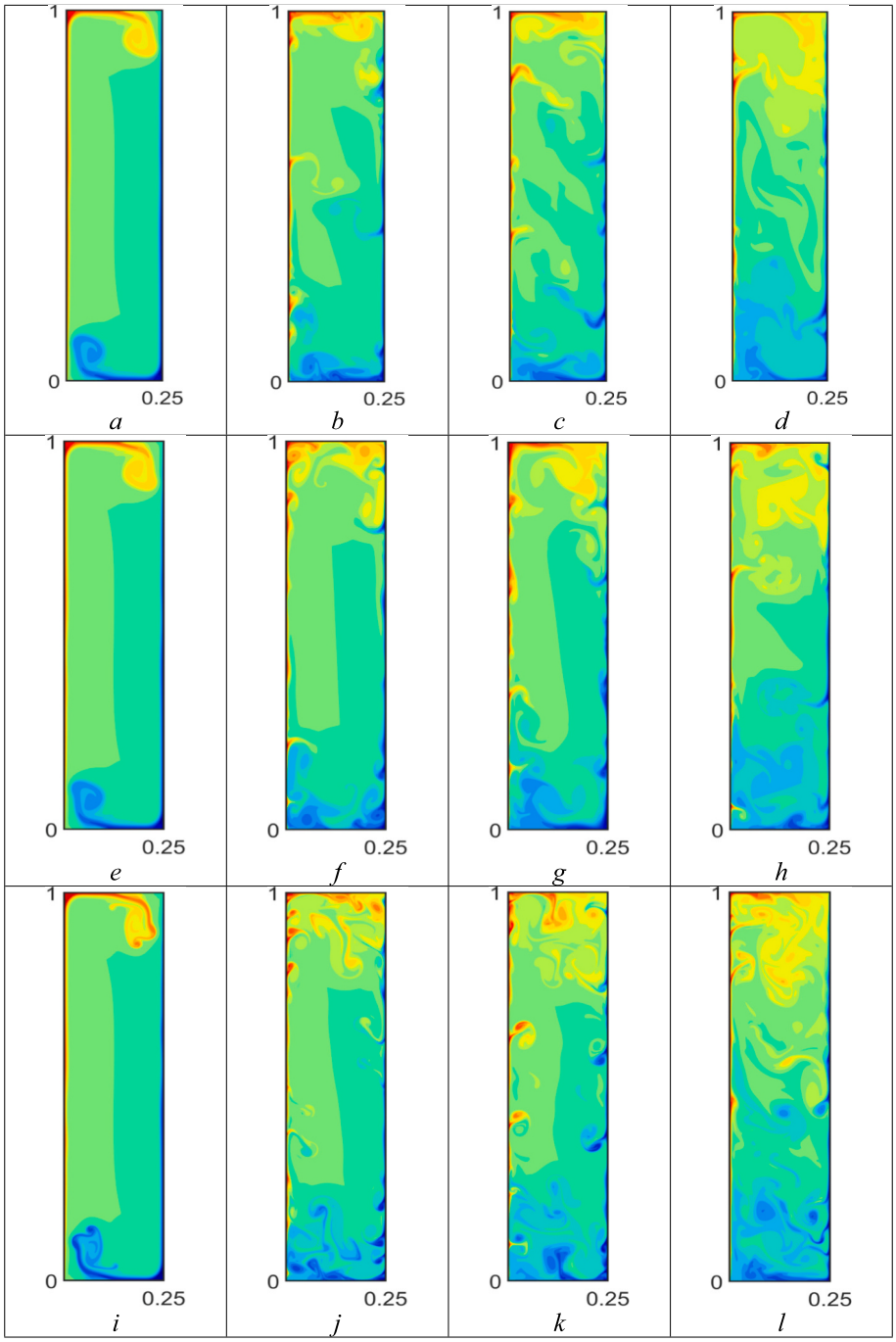


Fig. 16. Thermal fields computed by the first (a, b, c, d), second (e, f, g, h) and fourth (i, j, k, l) order finite difference solution of energy equation when $Ra = 10^{10}$: a, e,i) $\tau = 4.5$; b, f,j) $\tau = 9$; c, g,k) $\tau = 11.25$; d, h, l) $\tau = 15$.

Table 5
Numerical stability.

Ra	Grid	ω	1st order	2nd order	4th order
10^{10}	261 × 66	1.9664	1449*	3310*	stable
10^{10}	401 × 101	1.9449	stable	stable	stable
10^{10}	501 × 126	1.9387	5575*	stable	stable
10^{11}	1361 × 341	1.9709	29085*	stable	stable
10^{11}	1721 × 431	1.972	35531*	38293*	stable
10^{11}	2021 × 506	1.9692	69713*	56509*	stable
10^{12}	3201 × 801	1.9795	45858*	27373*	stable

* Unstable after this number of iterations.

In this regard, it was previously found [25] that the RLB based hybrid model reproduces very similar to [39] temperature contour in the square air-filled cavity when $Ra = 10^{10}$. It needs to be noted that the discrete unified gas kinetic scheme with the grid refinement was used in [39].

3.5. Numerical diffusion

The effect of the accuracy order of the governing equation on thermal patterns under transitional and turbulent flow behavior is analyzed via temperature contours dynamics. Thermal fields with the Rayleigh number corresponding to transitional zone are presented in Fig. 14.

It is found that all hybrid models under consideration reproduce the same thermal behavior. Single plumes formed along the isothermal

boundaries advance in the upstream and downstream directions reaching the opposite vertical wall with $\tau \leq 4$. However, it is clearly seen that the BGK-1 and TRT-2 models smear the plumes to some extent. On the other hand, the RLB-4 reproduces a distinct form of the thermal plume. When $\tau \geq 4$, different convective plumes behavior are predicted by the models under study. As could be assumed, the BGK-1 scheme significantly underestimates the production rate of thermal plumes. Along with that, it smears the steep temperature gradients. A slightly better performance is observed when using the TRT-2 model. This numerical coupling better predicts convective plumes generation. Finally, all relevant scales are resolved by the RLB-4 scheme. This model accurately simulates the evolution of thermal plumes with much better resolution of temperature gradients. Thermal dynamics under turbulent flow behavior is given in Fig. 15.

When analyzing the temperature contours, it is found that both the TRT-2 and RLB-4 models predict a similar thermal behavior before $\tau = 6$. Hereafter, the TRT-2 scheme significantly underestimates the rate of thermal plumes production. Along with that, this scheme smears temperature gradients as in the previous case with the $Ra = 2{\cdot}10^9$. On the other hand, the RLB-4 model accurately captures thermal plumes dynamics. Moreover, this model better resolves steep temperature gradients.

To the author knowledge, almost no attention is given to the method of energy equation solution under the hybrid lattice Boltzmann formulation. In order to fill this gap, the high order regularized lattice Boltzmann equations were coupled with the first, second and fourth order finite difference solution of energy equation. Numerical results are given in Fig. 16.

In general, numerical results are similar to those when comparing the BGK-1, TRT-2 and RLB-4 models. The order of approximation of energy equation affects the thermal plumes dynamics. The most dissipative first order scheme significantly smears temperature gradients and underestimates the rate of convective plumes generation along the isothermal walls. Similar thermal patterns are predicted when using the second order finite difference solution of energy equation. However, this finite difference scheme is able to catch more thermal plumes. Finally, the fourth order Runge-Kutta scheme accurately captures thermal plumes dynamics. Moreover, this scheme less smears temperature gradients in comparison with the first and second order schemes as could be expected.

Numerical stability is another important issue which has to be carefully treated within the lattice Boltzmann method. When using the hybrid formulation, the effect of the approximation order of the energy equation on numerical stability is of particular interest. Table 5 presents the results of numerical stability analysis when coupling the high order regularized lattice Boltzmann equations with the first, second and fourth order finite difference solution of energy equation.

It should be noted that the total number of iteration was set to 10^5. Numerical experiments were performed for an air-filled rectangular cavity with aspect ratio of 4. It is found that the approximation order of the energy equation significantly affects the numerical stability under turbulent flow behavior. This finding is associated with the numerical diffusion of the finite difference schemes. In other words, algorithm numerical stability of hybrid lattice Boltzmann models is affected by both velocity and temperature steep gradients. The RLB-4 model accurately resolves steep temperature gradient. As a result, this model has a very good or even outstanding numerical stability in some cases. In particular, the simulation is stable with the grid size and Rayleigh number of 261×66 and 10^{10}, respectively, whereas the other numerical schemes fail. The first order accurate scheme for energy equation has predictably the worst stability due to high level of smearing of temperature gradients. On the contrary, the finite difference solution of energy equation by the second order scheme has a good stability with moderate grid sizes and $Ra = 10^{10}$.

4. Summary

A comprehensive study on thermal convection simulation by means of the different hybrid lattice Boltzmann models was performed. The Bhatnagar-Gross-Krook, two relaxation time and high order regularized collision models coupled with the first, second and fourth order approximation of energy equation were considered. The Rayleigh number range of $10^4 \leq Ra \leq 10^{12}$ corresponding to laminar, laminar-to-turbulent and developed turbulent flows was analyzed.

As could be predicted, all modes under study reproduce the same results under laminar flow behavior. Local heat transfer characteristics computed by the BGK-1, TRT-2 and RLB-4 models are perfectly matched with each other. Moreover, deviation in integral characteristics does not exceed 3 % when comparing it with benchmark data of other researchers.

The main attention was given to laminar-to-turbulent and developed turbulent flows. Predictably, the worst numerical performance was observed when using BGK-1 model. It was found that this model can be applied only at the transitional flow behavior with $Ra \leq 6.4{\cdot}10^8$. The BGK-1 scheme underestimates the heat transfer rate by 9.4 % when $Ra = 2{\cdot}10^9$ which corresponds to low level turbulence. On the other hand, the TRT-2 model can deal with turbulent natural convection in a range of $6.4{\cdot}10^8 \leq Ra \leq 10^{10}$. The first and the second order statistics were in a satisfactory agreement with reference data. However, an error in the mean Nusselt number of 5.8 % was found when $Ra = 10^{10}$. Finally, the RLB-4 model demonstrated a very good performance in developed turbulence area. This numerical scheme predicts well the local, mean and turbulent characteristics of the buoyancy driven flow. Moreover, it was found that high order regularized lattice Boltzmann scheme coupled with the fourth order Runge-Kutta scheme is stable even with $Ra = 10^{12}$. In this case, the grid size and relaxation frequency were 801×3201 and 1.9795, respectively.

As could be expected, numerical diffusion significantly affects the thermal patterns under transitional and turbulent flow behavior. The BGK-1 and TRT-2 models smear the temperature gradients and underestimate the rate of thermal plumes generation along the isothermal walls. On the other hand, the RLB-4 scheme resolves all relevant scales. Moreover, this model has superior stability when coupling the high order regularized lattice Boltzmann equations with different numerical schemes for energy equation. In particular, the RLB-4 model was stable with the grid size and Rayleigh number of 261×66 and 10^{10}, respectively. However, this model was found the most computationally expensive as could be predicted.

The future outlook will include extension of the RLB-4 model to study turbulent natural convection coupled with radiation and conjugate heat transfer. Along with that, higher values of the Rayleigh number are of paramount interest. In the present study, the Ra was limited by the amount of a random access memory (RAM) of graphics processing unit (GPU NVIDIA GTX 1060 6 GB). Hence, it is possible that the RLB-4 model will be stable even with $Ra \geq 10^{13}$ when setting appropriate grid size. It will be found out in future works.

CRediT authorship contribution statement

Alexander Nee: Writing – review & editing, Writing – original draft.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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